

Лабораторная работа №14

Партиции, файловые системы, монтирование

Наурузова Айшат Магомедовна

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Российский университет дружбы народов, Москва, Россия

Цель работы

Получить навыки создания разделов на диске, форматирования файловых систем и их монтирования в Linux.

Ход выполнения работы

Создание разделов MBR с помощью fdisk

```
amnauruzova@amnauruzova:~$ su
Password:
root@amnauruzova:/home/amnauruzova#
root@amnauruzova:/home/amnauruzova# fdisk -l
Disk /dev/sda: 40 GiB, 42949672960 bytes, 83886080 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: gpt
Disk identifier: 0C51AD93-7E75-40C6-B42A-F80FD6565EFD
```

Device	Start	End	Sectors	Size	Type
/dev/sda1	2048	4095	2048	1M	BIOS boot
/dev/sda2	4096	2101247	2097152	1G	Linux extended boot
/dev/sda3	2101248	83884031	81782784	39G	Linux LVM


```
Disk /dev/sdb: 1.5 GiB, 1610612736 bytes, 3145728 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
```



```
Disk /dev/sdc: 1.5 GiB, 1610612736 bytes, 3145728 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
```

Создание основного раздела

```
Command (m for help):  
  
Command (m for help): n  
Partition type  
   p   primary (0 primary, 0 extended, 4 free)  
   e   extended (container for logical partitions)  
Select (default p): p  
Partition number (1-4, default 1):  
First sector (2048-3145727, default 2048):  
Last sector, +/-sectors or +/-size{K,M,G,T,P} (2048-3145727, default 3145727): +300M  
  
Created a new partition 1 of type 'Linux' and of size 300 MiB.  
  
Command (m for help): w  
The partition table has been altered.  
Calling ioctl() to re-read partition table.  
Syncing disks.  
  
root@amnauruzova:/home/amnauruzova#
```

Рис. 2: Создание основного раздела

Создание расширенного и логического разделов

```
root@amnauruzova:/home/amnauruzova# fdisk /dev/sdb

Welcome to fdisk (util-linux 2.40.2).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.


Command (m for help): n
Partition type
   p   primary (1 primary, 0 extended, 3 free)
   e   extended (container for logical partitions)
Select (default p): e
Partition number (2-4, default 2):
First sector (616448-3145727, default 616448):
Last sector, +/-sectors or +/-size{K,M,G,T,P} (616448-3145727, default 3145727):

Created a new partition 2 of type 'Extended' and of size 1.2 GiB.


Command (m for help): n
All space for primary partitions is in use.
Adding logical partition 5
First sector (618496-3145727, default 618496):
Last sector, +/-sectors or +/-size{K,M,G,T,P} (618496-3145727, default 3145727): +300M

Created a new partition 5 of type 'Linux' and of size 300 MiB.


Command (m for help): w
The partition table has been altered.
Calling ioctl() to re-read partition table.
Syncing disks.

root@amnauruzova:/home/amnauruzova#
```

Создание раздела подкачки

```
root@amnauruzova:/home/amnauruzova#  
root@amnauruzova:/home/amnauruzova# fdisk /dev/sdb  
  
Welcome to fdisk (util-linux 2.40.2).  
Changes will remain in memory only, until you decide to write them.  
Be careful before using the write command.  
  
Command (m for help): n  
All space for primary partitions is in use.  
Adding logical partition 6  
First sector (1234944-3145727, default 1234944):  
Last sector, +/-sectors or +/-size{K,M,G,T,P} (1234944-3145727, default 3145727): +300M  
  
Created a new partition 6 of type 'Linux' and of size 300 MiB.  
  
Command (m for help): t  
Partition number (1,2,5,6, default 6):  
Hex code or alias (type L to list all): 82  
  
Changed type of partition 'Linux' to 'Linux swap / Solaris'.  
  
Command (m for help): w  
The partition table has been altered.  
Calling ioctl() to re-read partition table.  
Syncing disks.  
  
root@amnauruzova:/home/amnauruzova# partprobe /dev/sdb  
root@amnauruzova:/home/amnauruzova#
```

Рис. 4: Создание раздела подкачки

Проверка и активация swar-раздела

```
8      0  41943040 sda
8      1    1024 sda1
8      2  1048576 sda2
8      3 40891392 sda3
8     16 1572864 sdb
8     17 307200 sdb1
8     18      0 sdb2
8     21 307200 sdb5
8     22 307200 sdb6
8     32 1572864 sdc
253    0 36753408 dm-0
253    1 4136960 dm-1
```

```
root@amnauruzova:/home/amnauruzova# fdisk /dev/sdb -l
```

Disk /dev/sdb: 1.5 GiB, 1610612736 bytes, 3145728 sectors

Disk model: VBOX HARDDISK

Units: sectors of 1 * 512 = 512 bytes

Sector size (logical/physical): 512 bytes / 512 bytes

I/O size (minimum/optimal): 512 bytes / 512 bytes

Disklabel type: dos

Disk identifier: 0x231de7c6

Device	Boot	Start	End	Sectors	Size	Id	Type
/dev/sdb1		2048	616447	614400	300M	83	Linux
/dev/sdb2		616448	3145727	2529280	1.2G	5	Extended
/dev/sdb5		618496	1232895	614400	300M	83	Linux
/dev/sdb6		1234944	1849343	614400	300M	82	Linux swap / Solaris

```
root@amnauruzova:/home/amnauruzova# mkswap /dev/sdb6
```

Setting up swapspace version 1, size = 300 MiB (314568704 bytes)

no label, UUID=590f2722-19f5-4ebb-bf2b-c1734f5e0fff

```
root@amnauruzova:/home/amnauruzova# swapon /dev/sdb6
```

```
root@amnauruzova:/home/amnauruzova# free -m
```

	total	used	free	shared	buff/cache	available
Mem:	3652	1311	1258	19	1324	2341
Swap:	4339	0	4339			

```
root@amnauruzova:/home/amnauruzova#
```

Создание GPT-разделов с помощью gdisk

Creating new GPT entries in memory.

Command (? for help): n
Partition number (1-128, default 1):
First sector (34-3145694, default = 2048) or {+-}size{KMGT}:
Last sector (2048-3145694, default = 3143679) or {+-}size{KMGT}: +300M
Current type is 8300 (Linux filesystem)
Hex code or GUID (L to show codes, Enter = 8300):
Changed type of partition to 'Linux filesystem'

Command (? for help): p
Disk /dev/sdc: 3145728 sectors, 1.5 GiB
Model: VBOX HARDDISK
Sector size (logical/physical): 512/512 bytes
Disk identifier (GUID): 24AEB274-5317-4167-92CE-5AB7D60D1F9E
Partition table holds up to 128 entries
Main partition table begins at sector 2 and ends at sector 33
First usable sector is 34, last usable sector is 3145694
Partitions will be aligned on 2048-sector boundaries
Total free space is 2531261 sectors (1.2 GiB)

Number	Start (sector)	End (sector)	Size	Code	Name
1	2048	616447	300.0 MiB	8300	Linux filesystem

Command (? for help): w

Final checks complete. About to write GPT data. THIS WILL OVERWRITE EXISTING PARTITIONS!!

Do you want to proceed? (Y/N): Y
OK; writing new GUID partition table (GPT) to /dev/sdc.
The operation has completed successfully.

root@amnauruzova:/home/amnauruzova#

Проверка созданных разделов GPT

```
8      1      1024 sda1
8      2    1048576 sda2
8      3   40891392 sda3
8     16    1572864 sdb
8     17    307200 sdb1
8     18         0 sdb2
8     21    307200 sdb5
8     22    307200 sdb6
8     32    1572864 sdc
8     33    307200 sdc1
253     0  36753408 dm-0
253     1   4136960 dm-1
```

```
root@amnauruzova:/home/amnauruzova# gdisk /dev/sdc -l
```

```
GPT fdisk (gdisk) version 1.0.10
```

```
Partition table scan:
```

```
MBR: protective
```

```
BSD: not present
```

```
APM: not present
```

```
GPT: present
```

```
Found valid GPT with protective MBR; using GPT.
```

```
Disk /dev/sdc: 3145728 sectors, 1.5 GiB
```

```
Model: VBOX HARDDISK
```

```
Sector size (logical/physical): 512/512 bytes
```

```
Disk identifier (GUID): 24AEB274-5317-4167-92CE-5AB7D60D1F9E
```

```
Partition table holds up to 128 entries
```

```
Main partition table begins at sector 2 and ends at sector 33
```

```
First usable sector is 34, last usable sector is 3145694
```

```
Partitions will be aligned on 2048-sector boundaries
```

```
Total free space is 2531261 sectors (1.2 GiB)
```

Number	Start (sector)	End (sector)	Size	Code	Name
1	2048	616447	300.0 MiB	8300	Linux filesystem

```
root@amnauruzova:/home/amnauruzova#
```

Форматирование разделов XFS и EXT4

```
root@amnauruzova:/home/amnauruzova# mkfs.xfs /dev/sdb1
meta-data=/dev/sdb1            isize=512    agcount=4, agsize=19200 blks
        =                       sectsz=512   attr=2, projid32bit=1
        =                       crc=1        finobt=1, sparse=1, rmapbt=1
        =                       reflink=1     bigtime=1 inobtcount=1 nrext64=1
        =                       exchange=0
data      =                       bsize=4096   blocks=76800, imaxpct=25
        =                       sunit=0       swidth=0 blks
naming    =version 2           bsize=4096   ascii-ci=0, ftype=1, parent=0
log        =internal log       bsize=4096   blocks=16384, version=2
        =                       sectsz=512   sunit=0 blks, lazy-count=1
realtime   =none               extsz=4096   blocks=0, rtextents=0
root@amnauruzova:/home/amnauruzova# xfs_admin -L xfsdisk /dev/sdb1
writing all SBs
new label = "xfsdisk"
root@amnauruzova:/home/amnauruzova#
root@amnauruzova:/home/amnauruzova# mkfs.ext4 /dev/sdb5
mke2fs 1.47.1 (20-May-2024)
Creating filesystem with 307200 1k blocks and 76912 inodes
Filesystem UUID: d86ecec-aa6a-4a96-9676-0fa10342aa30
Superblock backups stored on blocks:
    8193, 24577, 40961, 57345, 73729, 204801, 221185

Allocating group tables: done
Writing inode tables: done
Creating journal (8192 blocks): done
Writing superblocks and filesystem accounting information: done

root@amnauruzova:/home/amnauruzova# tune2fs -L ext4disk /dev/sdb5
bash: tune2fs: command not found...
root@amnauruzova:/home/amnauruzova# tune2fs -L ext4disk /dev/sdb5
tune2fs 1.47.1 (20-May-2024)
root@amnauruzova:/home/amnauruzova# tune2fs -o acl,user_xattr /dev/sdb5
tune2fs 1.47.1 (20-May-2024)
Invalid mount option set: acl,user_xattr
root@amnauruzova:/home/amnauruzova# tune2fs -o acl,user_xattr /dev/sdb5
tune2fs 1.47.1 (20-May-2024)
root@amnauruzova:/home/amnauruzova#
```

```
root@amnauruzova:/home/amnauruzova#  
root@amnauruzova:/home/amnauruzova# mkdir /mnt/tmp  
root@amnauruzova:/home/amnauruzova# mount /dev/sdb5 /mnt/tmp  
root@amnauruzova:/home/amnauruzova# mount | grep mnt  
/dev/sdb5 on /mnt/tmp type ext4 (rw,relatime,seclabel)  
root@amnauruzova:/home/amnauruzova# umount /dev/sdb5  
root@amnauruzova:/home/amnauruzova# mount | grep mnt  
root@amnauruzova:/home/amnauruzova#
```

Рис. 9: Ручное монтирование и отмонтирование

Монтирование раздела XFS через /etc/fstab

```
#
# /etc/fstab
# Created by anaconda on Tue Sep  2 12:53:30 2025
#
# Accessible filesystems, by reference, are maintained under '/dev/disk/'.
# See man pages fstab(5), findfs(8), mount(8) and/or blkid(8) for more info.
#
# After editing this file, run 'systemctl daemon-reload' to update systemd
# units generated from this file.
#
UUID=34884054-27f2-4624-8671-1f66bf893ae2 / xfs defaults 0 0
UUID=5bb25999-de96-4a24-930c-75854bf11fc1 /boot xfs defaults 0 0
UUID=774abe3d-c83f-4c4f-90c6-404f43a900ee none swap defaults 0 0
UUID=e97bbe69-dd87-4579-880c-1ca10bbb317 /mnt/data xfs defaults 1 2
```

Рис. 10: Добавление записи в /etc/fstab

Проверка монтирования через fstab

```
root@amnauruzova:/home/amnauruzova# nano /etc/fstab
root@amnauruzova:/home/amnauruzova# mount -a
mount: (hint) your fstab has been modified, but systemd still uses
the old version; use 'systemctl daemon-reload' to reload.
root@amnauruzova:/home/amnauruzova# df -h
```

Filesystem	Size	Used	Avail	Use%	Mounted on
/dev/mapper/rl_vbox-root	35G	6.1G	29G	18%	/
devtmpfs	4.0M	0	4.0M	0%	/dev
tmpfs	1.8G	84K	1.8G	1%	/dev/shm
tmpfs	731M	11M	721M	2%	/run
tmpfs	1.0M	0	1.0M	0%	/run/credentials/systemd-journald.service
/dev/sda2	960M	377M	584M	40%	/boot
tmpfs	366M	144K	366M	1%	/run/user/1000
tmpfs	366M	60K	366M	1%	/run/user/0
/dev/sdb1	236M	20M	217M	9%	/mnt/data

```
root@amnauruzova:/home/amnauruzova#
```

Рис. 11: Проверка монтирования через fstab

```
Command (? for help): n
Partition number (2-128, default 2):
First sector (34-3145694, default = 616448) or {+-}size{KMGTp}:
Last sector (616448-3145694, default = 3143679) or {+-}size{KMGTp}: +300M
Current type is 8300 (Linux filesystem)
Hex code or GUID (L to show codes, Enter = 8300):
Changed type of partition to 'Linux filesystem'

Command (? for help): n
Partition number (3-128, default 3):
First sector (34-3145694, default = 1230848) or {+-}size{KMGTp}:
Last sector (1230848-3145694, default = 3143679) or {+-}size{KMGTp}: +300M
Current type is 8300 (Linux filesystem)
Hex code or GUID (L to show codes, Enter = 8300): 8200
Changed type of partition to 'Linux swap'

Command (? for help): p
Disk /dev/sdc: 3145728 sectors, 1.5 GiB
Model: VBOX HARDDISK
Sector size (logical/physical): 512/512 bytes
Disk identifier (GUID): 24AEB274-5317-4167-92CE-5AB7D60D1F9E
Partition table holds up to 128 entries
Main partition table begins at sector 2 and ends at sector 33
First usable sector is 34, last usable sector is 3145694
Partitions will be aligned on 2048-sector boundaries
Total free space is 1302461 sectors (636.0 MiB)

Number  Start (sector)    End (sector)  Size      Code  Name
   1            2048             616447     300.0 MiB   8300   Linux filesystem
   2          616448          1230847     300.0 MiB   8300   Linux filesystem
   3          1230848          1845247     300.0 MiB   8200   Linux swap

Command (? for help): w

Final checks complete. About to write GPT data. THIS WILL OVERWRITE EXISTING
PARTITIONS!!

Do you want to proceed? (Y/N): Y
OK; writing new GUID partition table (GPT) to /dev/sdc.
The operation has completed successfully.
root@omniuxuser: /home/omniuxuser#
```


Форматирование и настройка swar

```
root@amnauruzova:/home/amnauruzova# cat /proc/partitions
major minor #blocks name

11        0      1048575 sr0
 8         0    41943040 sda
 8         1       1024 sda1
 8         2    1048576 sda2
 8         3   40891392 sda3
 8        16    1572864 sdb
 8        17    307200 sdb1
 8        18         0 sdb2
 8        21    307200 sdb5
 8        22    307200 sdb6
 8        32    1572864 sdc
 8        33    307200 sdc1
 8        34    307200 sdc2
 8        35    307200 sdc3
253        0   36753408 dm-0
253        1    4136960 dm-1

root@amnauruzova:/home/amnauruzova#
root@amnauruzova:/home/amnauruzova# mkfs.ext4 /dev/sdc2
mke2fs 1.47.1 (20-May-2024)
Creating filesystem with 307200 1k blocks and 76912 inodes
Filesystem UUID: ef6eb4a8-920f-4d8b-96f8-b4e9a925c30b
Superblock backups stored on blocks:
    8193, 24577, 40961, 57345, 73729, 204801, 221185

Allocating group tables: done
Writing inode tables: done
Creating journal (8192 blocks): done
Writing superblocks and filesystem accounting information: done

root@amnauruzova:/home/amnauruzova# tune2fs -L ext4disk2 /dev/sdc2
tune2fs 1.47.1 (20-May-2024)
root@amnauruzova:/home/amnauruzova# tune2fs -o acl,user_xattr /dev/sdc2
tune2fs 1.47.1 (20-May-2024)
root@amnauruzova:/home/amnauruzova# mkswap /dev/sdc3
Setting up swapspace version 1, size = 300 MiB (314568704 bytes)
no label, UUID=6c12fcce-02a4-4084-8214-c952a4d0cbda
```

Добавление разделов в /etc/fstab

```
#
# /etc/fstab
# Created by anaconda on Tue Sep  2 12:53:30 2025
#
# Accessible filesystems, by reference, are maintained under '/dev/disk/'.
# See man pages fstab(5), findfs(8), mount(8) and/or blkid(8) for more info.
#
# After editing this file, run 'systemctl daemon-reload' to update systemd
# units generated from this file.
#
UUID=34884054-27f2-4624-8671-1f66bf893ae2 / xfs defaults 0 0
UUID=5bb25999-de96-4a24-930c-75854bf11fc1 /boot xfs defaults 0 0
UUID=774abe3d-c83f-4c4f-90c6-404f43a900ee none swap defaults 0 0
UUID=e97bbe69-dd87-4579-880c-1ca10bbb317 /mnt/data xfs defaults 1 2
UUID=ef6eb4a8-920f-4d8b-96f8-b4e9a925c30b /mnt/data-ext ext4 defaults 1 2
UUID=6c12fcce-02a4-4084-8214-c952a4d0cbda none swap defaults 0 0
```

Рис. 14: Редактирование файла /etc/fstab

Проверка монтирования и swap

```
amnauruzova@amnauruzova:~$  
amnauruzova@amnauruzova:~$ df -h  
Filesystem      Size  Used Avail Use% Mounted on  
/dev/mapper/rl_vbox-root 35G  6.1G   29G  18% /  
devtmpfs         4.0M    0  4.0M   0% /dev  
tmpfs            1.8G   84K   1.8G   1% /dev/shm  
tmpfs            731M   9.3M  722M   2% /run  
tmpfs            1.0M    0   1.0M   0% /run/credentials/systemd-journald.service  
/dev/sda2        960M  377M  584M  40% /boot  
/dev/sdb1        236M   20M  217M   9% /mnt/data  
/dev/sdc2        272M   14K  253M   1% /mnt/data-ext  
tmpfs            366M  144K  366M   1% /run/user/1000  
amnauruzova@amnauruzova:~$ mount -a | grep mnt  
amnauruzova@amnauruzova:~$ mount | grep mnt  
/dev/sdb1 on /mnt/data type xfs (rw,relatime,seclabel,attr2,inode64,logbufs=8,logbsize=32k,noquota)  
/dev/sdc2 on /mnt/data-ext type ext4 (rw,relatime,seclabel)  
amnauruzova@amnauruzova:~$ free -m  
              total        used        free      shared  buff/cache   available  
Mem:           3652         1245         1973          18          668         2407  
Swap:          4339              0         4339
```

Рис. 15: Проверка монтирования и swap

Итоги работы

В ходе лабораторной работы были изучены утилиты **fdisk** и **gdisk**, созданы разделы с разметкой **MBR** и **GPT**, отформатированы в **XFS** и **EXT4**, а также настроен и активирован раздел подкачки **swap**.

Освоены команды **mkfs**, **mkswap**, **mount**, **blkid**, и механизм автоматического монтирования через файл **/etc/fstab**.